QUASI-CRYSTALS ARE THE $q = 0$ AVATAR OF QSYM

RICK (PROVE 277, 2026-11-06)

Abstract. A self-contained account of type-A quasi-crystals in the sense of Cain–Malheiro and Cain–Malheiro–Rodrigues–Rodrigues, written so that no idea is used before it is proved. The whole theory runs on one identity. Define the defect $\delta_i = \mathrm{wt}_{i+1} - \tilde{\varepsilon}_i$, which counts i-brackets. Then

$$\delta_i(x \otimes y) = \delta_i(x) + \delta_i(y) + \min(\tilde{\varepsilon}_i(x), \tilde{\varphi}_i(y)).$$

From it we get, in order: the quasi-tensor rule (signature rule plus kill clause, owned by Cain–Guilherme–Malheiro); strict monoidality of quasification; components as standardisation classes with character F_α (Gessel); K_0 of type-A quasi-crystals $\cong \mathrm{QSym}_n$ as rings; Levi branching inducing Δ; and $\mathrm{Sym}_n \hookrightarrow \mathrm{QSym}_n$ as the shadow of "a crystal is a union of quasi-crystals". The punchline is that the crystal tensor product shadows the *product* and branching shadows the *coproduct*. They agree on Sym only because Sym is self-dual, and quasi-crystals are where they come apart. The content of every result is folklore at the level of Gessel 1984 and Malvenuto–Reutenauer 1995. The packaging is ours, and we did not find it stated in the sources listed in §10.

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1. Conventions and black boxes

Type A_{n-1}. Weights live in $\mathbb{Z}^n$, $\alpha_i = e_i - e_{i+1}$, $\langle \lambda, \alpha_i \rangle = \lambda_i - \lambda_{i+1}$, and $i \in [n-1]$. Words $w = w_1 \cdots w_N \in [n]^N$ have weight $\mathrm{wt}(w) = (\#1, \ldots, \#n)$ and monomial $x^w = x^{\mathrm{wt}w}$. $[n]^0 = \{\varnothing\}$.

Word order. Throughout, the tensor product $x \otimes y$ corresponds to the *concatenation* xy. This is the opposite of Kashiwara's original convention and of the convention of Cain–Guilherme–Malheiro [CGM, arXiv:2301.00271, Def. 6.9]. The bridge is the sentence in [CMRR2, arXiv:2309.14898, p. 10]: "$x \ddot{\otimes} y$ is identified with the word yx". Anything quoted from CGM must therefore be word-reversed. Since the kill clause (Theorem 4) is asymmetric, this is not cosmetic.

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The following are used without proof. They appear only in §8 and in remarks, and nothing before §8 depends on them.

Black box 1. For seminormal crystals B, B', the tensor product of Definition 2.4 is a seminormal crystal (Kashiwara 1991; Bump–Schilling, *Crystal Bases*, Ch. 2). For word crystals this is Lemma 2.5 below, and only there is it needed before §8.

Black box 2. Every irreducible polynomial $\mathfrak{gl}_n$ crystal $B(\lambda)$ ($\ell(\lambda) \le n$) is isomorphic to a connected component of the word crystal $[n]^{|\lambda|}$ (column/row reading word; Kashiwara–Nakashima, Bump–Schilling Ch. 3, 8). Every connected component of $[n]^N$ is some $B(\lambda)$, and $\operatorname{ch} B(\lambda) = s_\lambda(x_1, \dots, x_n)$.

Black box 3. Gessel: $s_\lambda = \sum_{T \in SYT(\lambda)} F_{\mathrm{Des}(T)}$ [Stanley EC2, Thm 7.19.7]. This is used only to make Corollary 8.1(c) explicit. The ring-map statement does not need it.

2. Words, brackets, crystals

Definition 2.1 (i-brackets)**.** Fix i. In a word w, view each letter $i+1$ as "(" and each letter i as ")". Ignore all other letters. Match parentheses in the usual way, by repeatedly matching an adjacent pair "()" in the $\{i, i+1\}$-subword and deleting it. The result is independent of the order of deletions; it is the usual matching. The matched pairs are the *i-brackets* of w. After deleting them, the reduced subword is $i^{\tilde\varphi}(i+1)^{\tilde\varepsilon}$, and

$$\tilde\varepsilon_i(w) = \#\text{unmatched } (i+1), \qquad \tilde\varphi_i(w) = \#\text{unmatched } i.$$

$\tilde e_i(w)$ changes the *leftmost* unmatched $i{+}1$ to i (it is 0 if $\tilde\varepsilon_i(w) = 0$). $\tilde f_i(w)$ changes the *rightmost* unmatched i to $i+1$ (it is 0 if $\tilde\varphi_i(w) = 0$).

Example 2.2. $w = 2\,1\,1$, $i = 1$: the 2 matches the first 1, leaving one unmatched 1. So $\tilde\varepsilon_1 = 0$, $\tilde\varphi_1 = 1$, and $\tilde f_1(211) = 212$. In 212 the first 2 matches the 1, and the last 2 is unmatched, so $\tilde e_1(212) = 211$.

Definition 2.3 (seminormal crystal)**.** A *seminormal $\mathfrak{gl}_n$ crystal* is a finite set B with $\mathrm{wt} : B \to \mathbb{Z}^n$ and partial maps $\tilde e_i, \tilde f_i : B \to B \sqcup \{0\}$ such that $\tilde f_i x = y \iff \tilde e_i y = x$, $\mathrm{wt}(\tilde e_i x) = \mathrm{wt}(x) + \alpha_i$, and $\tilde\varepsilon_i(x) = \max\{k : \tilde e_i^k x \neq 0\}$, $\tilde\varphi_i(x) = \max\{k : \tilde f_i^k x \neq 0\}$ satisfy $\tilde\varphi_i(x) - \tilde\varepsilon_i(x) = \langle \mathrm{wt} x, \alpha_i \rangle$.

The word crystal $[n]^N$ is seminormal: $\tilde e_i$ changes an $i+1$ into an i, and the i-strings are the standard $\{i, i+1\}$-strings of the reduced subword.

Definition 2.4 (tensor product, word order)**.** For seminormal B, B', let $B \otimes B' = B \times B'$ with additive weight. Write $\varepsilon = \tilde\varepsilon_i(x)$, $\varphi = \tilde\varphi_i(x)$, $\varepsilon' = \tilde\varepsilon_i(y)$, $\varphi' = \tilde\varphi_i(y)$, and $m = \min(\varepsilon, \varphi')$. Then

$$\tilde\varepsilon_i(x \otimes y) = \varepsilon + \varepsilon' - m, \quad \tilde\varphi_i(x \otimes y) = \varphi + \varphi' - m,$$

$\tilde e_i$ acts on x if $\varepsilon > \varphi'$ and on y otherwise, and $\tilde f_i$ acts on y if $\varphi' > \varepsilon$ and on x otherwise.

Lemma 2.5 (concatenation is the tensor product)**.** *The map $(u, v) \mapsto uv$ is an isomorphism $[n]^m \otimes [n]^k \to [n]^{m+k}$ of seminormal crystals. Moreover the number of i-brackets of uv is*

$$\#\textit{brackets}(u) + \#\textit{brackets}(v) + \min(\tilde\varepsilon_i(u), \tilde\varphi_i(v)).$$

Proof. First match brackets inside u and inside v. This leaves $i^{\tilde\varphi(u)}(i+1)^{\tilde\varepsilon(u)} \,\big|\, i^{\tilde\varphi(v)}(i+1)^{\tilde\varepsilon(v)}$. The only adjacent "()" pairs remaining straddle the bar: they match the $\tilde\varepsilon(u)$ trailing ('s of u against the $\tilde\varphi(v)$ leading)'s of v, which gives $m = \min(\tilde\varepsilon(u), \tilde\varphi(v))$ new brackets. The result is $i^{\tilde\varphi(u)+\tilde\varphi(v)-m}(i+1)^{\tilde\varepsilon(u)+\tilde\varepsilon(v)-m}$, which gives both $\tilde\varepsilon, \tilde\varphi$ formulas and the bracket count. The leftmost unmatched (lies in u iff $\tilde\varepsilon(u) - m > 0$, i.e. iff $\tilde\varepsilon(u) > \tilde\varphi(v)$. The rightmost unmatched) lies in v iff $\tilde\varphi(v) - m > 0$, i.e. iff $\tilde\varphi(v) > \tilde\varepsilon(u)$. Within the factor that is acted on, the unmatched

letter chosen is the one the factor's own operator chooses. Weight is additive. The map is a bijection because lengths are fixed. □

3. Quasi-crystals, quasification, defect

Definition 3.1 (word quasi-crystal; CMRR1 arXiv:2309.14887 Def. 2.13, word order)**.** A word w has an *i-inversion* if it has a subsequence $(i+1)\cdots i$. If it does not, its $\{i, i+1\}$-subword is literally $i^a(i+1)^b$, and we set

$$\ddot{\varepsilon}_i(w) = b, \quad \ddot{\varphi}_i(w) = a, \quad \ddot{e}_i(w) = \text{leftmost } (i+1) \mapsto i, \quad \ddot{f}_i(w) = \text{rightmost } i \mapsto i+1$$

($\perp$ if $b = 0$, resp. $a = 0$). If w has an i-inversion, then $\ddot{\varepsilon}_i = \ddot{\varphi}_i = +\infty$ and $\ddot{e}_i(w) = \ddot{f}_i(w) = \perp$.

Definition 3.2 (quasi-crystal, minimal version)**.** A *($\mathfrak{gl}_n$-)quasi-crystal* is a set with wt, partial maps $\ddot{e}_i, \ddot{f}_i$ with $\ddot{f}_i x = y \iff \ddot{e}_i y = x$ and $\mathrm{wt}(\ddot{e}_i x) = \mathrm{wt}\, x + \alpha_i$, and $\ddot{\varepsilon}_i, \ddot{\varphi}_i \in \mathbb{Z}_{\geq 0} \sqcup \{+\infty\}$. A *morphism* is a weight-preserving bijection that commutes with all of these data. (CMRR impose further axioms. We never need them, because all objects below come from words.)

Definition 3.3 (quasification; CMRR2 arXiv:2309.14898 Def. 4.5)**.** For a seminormal crystal B, Q_B has the same set and weights, and

$$\ddot{\varepsilon}_i(x) = \tilde{\varepsilon}_i(x),\ \ddot{e}_i(x) = \tilde{e}_i(x) \ \text{ if } \tilde{\varepsilon}_i(x) = \mathrm{wt}_{i+1}(x); \qquad \ddot{\varepsilon}_i(x) = \infty,\ \ddot{e}_i(x) = \perp \ \text{ otherwise.}$$

Also $\ddot{\varphi}_i := \ddot{\varepsilon}_i + \langle \mathrm{wt}, \alpha_i \rangle$ (with $\infty + c = \infty$) and $\ddot{f}_i x = y :\iff \ddot{e}_i y = x$. CMRR state this for connected Stembridge crystals. The rule is pointwise, so it applies verbatim to any seminormal crystal, componentwise.

Definition 3.4 (defect)**.** $\delta_i(x) := \mathrm{wt}_{i+1}(x) - \tilde{\varepsilon}_i(x)$. By seminormality $\delta_i(x) = \mathrm{wt}_i(x) - \tilde{\varphi}_i(x)$ as well. Call x *i-alive* if $\delta_i(x) = 0$, and *i-dead* otherwise. Thus Q_B keeps exactly the i-alive part of each i-string's structure and declares the rest dead.

Lemma 3.5. *Let B be seminormal.*

(a) δ_i is constant on i-strings.
(b) If every weight of B is polynomial ($\in \mathbb{Z}^n_{\geq 0}$), then $\delta_i \geq 0$ on B.
(c) On alive x: $\ddot{f}_i(x) = \tilde{f}_i(x)$, $\ddot{\varepsilon}_i = \tilde{\varepsilon}_i$, $\ddot{\varphi}_i = \tilde{\varphi}_i$.
(d) On words, $\delta_i(w)$ is the number of i-brackets of w, and $\delta_i(w) = 0$ iff w has no i-inversion.
(e) Hence $Q_{[n]^N}$ is the word quasi-crystal of Definition 3.1.

Proof. (a) $\tilde{e}_i$ lowers wt_{i+1} by 1 and lowers $\tilde{\varepsilon}_i$ by 1. (b) $y = \tilde{e}_i^{\tilde{\varepsilon}_i(x)} x$ exists, and $\mathrm{wt}_{i+1}(y) = \mathrm{wt}_{i+1}(x) - \tilde{\varepsilon}_i(x) = \delta_i(x)$. This is ≥ 0 by polynomiality. (c) If $\tilde{f}_i x = y$, then $\tilde{e}_i y = x$ and y is alive by (a), so $\ddot{e}_i y = x$, i.e. $\ddot{f}_i x = y$. Conversely, $\ddot{e}_i y = x$ forces $\tilde{e}_i y = x$, so $y = \tilde{f}_i x$. Finally $\ddot{\varphi}_i = \tilde{\varepsilon}_i + \langle \mathrm{wt}, \alpha_i \rangle = \tilde{\varphi}_i$. (d) $\#(i+1) = \#\text{matched} + \#\text{unmatched}\ (i+1)$, so $\delta_i(w)$ is the number of brackets. If w has an $i+1$ before an i, then between them the $\{i, i+1\}$-subword must contain an adjacent "( )", so there is a bracket. The converse is trivial. (e) By (d), alive means no i-inversion. In that case the subword is $i^a(i+1)^b$ with nothing matched, so $\tilde{e}_i$ changes the leftmost $i+1$, which is $\ddot{e}_i$ of Def. 3.1, and $\tilde{\varepsilon}_i = b$. $\ddot{f}_i$ then agrees by (c). □

Example 3.6 (the crystal edge that quasification deletes)**.** In $[2]^3$ the crystal component $\{211 \xrightarrow{1} 212\} \cong B(2,1)$ has $s_{21}(x_1, x_2) = x_1^2 x_2 + x_1 x_2^2$. Both words contain the bracket $2 \cdots 1$, so $\delta_1 = 1$ on both and both are 1-dead. Q deletes the edge and leaves two singletons, $\{211\}$ (with std = 312, character $x_1^2 x_2 = F_{21}(x_1, x_2)$) and $\{212\}$ (with std = 213, character $x_1 x_2^2 = F_{12}(x_1, x_2)$).

$$211 \xrightarrow{\tilde{f}_1} 212 \qquad\qquad 211 \xrightarrow{\text{dead}} 212$$

crystal $B(2,1)$ $\qquad\qquad$ Q: two components

Non-example. Q is not "delete edges between different standardisations" applied to an arbitrary crystal. On a non-polynomial crystal such as $\det^{-1}$ (one point, weight $(-1,-1)$, $\tilde{\varepsilon} = 0$), the point is dead ($\delta_1 = -1$) although it has no edge at all.

4. The defect identity and the quasi-tensor rule

Definition 4.1 (quasi-tensor product, word order)**.** For quasi-crystals Q, Q', let $Q \ddot{\otimes} Q' = Q \times Q'$ with additive weight, and for each i:

- $\ddot{\varepsilon}_i(x \ddot{\otimes} y) = \ddot{\varphi}_i(x \ddot{\otimes} y) = \infty$ (and $\ddot{e}_i = \ddot{f}_i = \bot$) iff $\ddot{\varepsilon}_i(x) = \infty$ or $\ddot{\varepsilon}_i(y) = \infty$ or $\big(\ddot{\varepsilon}_i(x) > 0$ and $\ddot{\varphi}_i(y) > 0\big)$. This last clause is the *kill clause*;
- otherwise $\ddot{\varepsilon}_i = \ddot{\varepsilon}_i(x) + \ddot{\varepsilon}_i(y)$ and $\ddot{\varphi}_i = \ddot{\varphi}_i(x) + \ddot{\varphi}_i(y)$. $\ddot{e}_i$ acts on x if $\ddot{\varepsilon}_i(x) > 0$, else on y if $\ddot{\varepsilon}_i(y) > 0$, else gives $\bot$. $\ddot{f}_i$ acts on y if $\ddot{\varphi}_i(y) > 0$, else on x if $\ddot{\varphi}_i(x) > 0$, else gives $\bot$.

This is CGM's quasi-tensor product [2301.00271 Thm 5.1, Prop. 5.9, Def. 6.9] with factors reversed (see §1).

Proposition 4.2 (defect identity)**.** *For seminormal B, B' and any weights,*

$$\boxed{\delta_i(x \otimes y) = \delta_i(x) + \delta_i(y) + \min\big(\tilde{\varepsilon}_i(x), \tilde{\varphi}_i(y)\big).}$$

On words, the three terms count the brackets inside u, the brackets inside v, and the brackets crossing the cut (Lemma 2.5).

Proof. wt_{i+1} is additive, and $\tilde{\varepsilon}_i(x \otimes y) = \tilde{\varepsilon}_i(x) + \tilde{\varepsilon}_i(y) - \min(\tilde{\varepsilon}_i(x), \tilde{\varphi}_i(y))$. Subtract. □

Theorem 4.3 (quasification is strict monoidal; Q66)**.** *Let B, B' be seminormal crystals whose weights are all polynomial. Then the identity of $B \times B'$ is an equality of quasi-crystals*

$$Q_{B \otimes B'} = Q_B \ddot{\otimes} Q_{B'}.$$

Proof. Fix i, $x \in B$, $y \in B'$, and write $\varepsilon = \tilde{\varepsilon}_i(x)$, $\varphi' = \tilde{\varphi}_i(y)$. By Lemma 3.5(b) all three terms of the defect identity are ≥ 0, so

$$x \otimes y \text{ alive} \iff \delta(x) = 0 \ \wedge\ \delta(y) = 0 \ \wedge\ \min(\varepsilon, \varphi') = 0.$$

Dead sets agree. If x is dead, then $\ddot{\varepsilon}(x) = \infty$ in Q_B and Def. 4.1 kills, as $Q_{B \otimes B'}$ does. The same holds for y. If both are alive, then $\ddot{\varepsilon}(x) = \varepsilon$ and $\ddot{\varphi}(y) = \varphi'$ by Lemma 3.5(c), and "$\min(\varepsilon, \varphi') > 0$" is exactly the kill clause. On the dead set both sides have $\ddot{\varepsilon} = \ddot{\varphi} = \infty$ and $\ddot{e} = \ddot{f} = \bot$. *Alive set.* Here $m = 0$, so $\tilde{\varepsilon}(x \otimes y) = \varepsilon + \tilde{\varepsilon}(y) = \ddot{\varepsilon}(x) + \ddot{\varepsilon}(y)$, and likewise for $\tilde{\varphi}$. The signature rule applies $\tilde{e}$ to x iff $\varepsilon > \varphi'$. Since $m = 0$, either $\varepsilon = 0$, in which case it acts on y (or both rules give $\bot$), or $\varphi' = 0 < \varepsilon$, in which case it acts on x. That is the $\ddot{\otimes}$ rule "on x if $\ddot{\varepsilon}(x) > 0$, else on y". On the left, $\ddot{e} = \tilde{e}$ because $x \otimes y$ is alive. On the right, $\ddot{e}$ applied to the alive factor is $\tilde{e}$. The $\ddot{f}$ case is symmetric, using Lemma 3.5(c). □

Theorem A (quasi-tensor rule on words; owner: Cain–Guilherme–Malheiro)**.** *For all words $u \in [n]^m$, $v \in [n]^k$, the map $(u, v) \mapsto uv$ is an isomorphism of word quasi-crystals $[n]^m \ddot{\otimes} [n]^k \cong [n]^{m+k}$. Explicitly: uv has an i-inversion iff u does, or v does, or ($\ddot{\varepsilon}_i(u) > 0$ and $\ddot{\varphi}_i(v) > 0$). Otherwise $\ddot{\varepsilon}, \ddot{\varphi}$ add, $\ddot{e}_i$ acts on the leftmost factor containing an $i{+}1$, and $\ddot{f}_i$ acts on the rightmost factor containing an i.*

Proof (as a corollary). Word crystals have polynomial weights. By Lemma 2.5, $[n]^{m+k} \cong [n]^m \otimes [n]^k$ as crystals, and quasification commutes with isomorphisms. Apply Theorem 4.3 and Lemma 3.5(e). □

Direct proof (PROVE 275). A subsequence $(i+1)\cdots i$ of uv lies in u, in v, or straddles the cut. For inversion-free u, v, straddling means u contains $i+1$ and v contains i, i.e. $\ddot{\varepsilon}(u) > 0$ and $\ddot{\varphi}(v) > 0$. Otherwise the subword is $i^{\ddot{\varphi}(u)}(i+1)^{\ddot{\varepsilon}(u)} i^{\ddot{\varphi}(v)}(i+1)^{\ddot{\varepsilon}(v)}$ with $\ddot{\varepsilon}(u)\ddot{\varphi}(v) = 0$, i.e. $i^{\ddot{\varphi}(u)+\ddot{\varphi}(v)}(i+1)^{\ddot{\varepsilon}(u)+\ddot{\varepsilon}(v)}$. Its leftmost $i+1$ is in u iff $\ddot{\varepsilon}(u) > 0$. □

Remark 4.4 (restating compresses). In PROVE 275, Theorem A was proved first and Q66 (PROVE 276) afterwards. In dependency order Theorem A is the special case $B = [n]^m$, $B' = [n]^k$ of Q66. The kill clause *is* the cross-cut term $\min(\tilde{\varepsilon}(x), \tilde{\varphi}(y))$ of the defect identity. So the quasi-tensor rule is the Kashiwara signature rule restricted to pairs with no bracket across the cut. The asymmetry of the kill clause ($\ddot{\varepsilon}(u) > 0, \ddot{\varphi}(v) > 0$ kills, while $\ddot{\varphi}(u) > 0, \ddot{\varepsilon}(v) > 0$ is harmless) is the asymmetry of bracketing: an $i+1$ matches a *later* i.

Example 4.5. $n = 2$, $i = 1$. Take $u = 2$, $v = 1$: $\ddot{\varepsilon}(u) = 1$ and $\ddot{\varphi}(v) = 1$, so the kill clause fires and 21 is dead (one bracket, crossing the cut). Take $u = 1$, $v = 2$: no kill, and 12 has $\ddot{\varepsilon} = \ddot{\varphi} = 1$, $\ddot{e}_1(12) = 11$, $\ddot{f}_1(12) = 22$.

Example 4.6 (sharpness: polynomiality is needed)**.** (i) $\mathfrak{gl}_2$, $B = \det^{-1}$ (one point, weight $(-1,-1)$) and $B' = \det$ (one point, weight $(1,1)$). Then $\tilde{\varepsilon} = \tilde{\varphi} = 0$, $\delta = -1$ and $+1$, so both points are dead, and $Q_B \ddot{\otimes} Q_{B'}$ has $\ddot{\varepsilon}_1 = \infty$. But $B \otimes B'$ has weight $(0,0)$, $\delta = 0$, so $Q_{B\otimes B'}$ has $\ddot{\varepsilon}_1 = 0$. The defects cancelled. (ii) A lost edge: $B = B(0,-1)$ (weights $(0,-1) \to (-1,0)$, $\delta \equiv -1$) and $B' = B(2,1)$ ($\delta \equiv +1$). Then $B \otimes B' \supset B(2,0)$, whose 3-element 1-string (weights $(2,0),(1,1),(0,2)$) is alive in $Q_{B\otimes B'}$ with $\ddot{\varepsilon}_1 = 0, 1, 2$. Every element of $Q_B \ddot{\otimes} Q_{B'}$ is dead.

5. Components are standardisation classes (Theorem B)

Definition 5.1. $\mathrm{std}(w) \in S_N$ labels the letters of w by $1, \ldots, N$, smallest letters first and equal letters left to right. For $\sigma \in S_N$ (as a word), $\mathrm{iDes}(\sigma) = \{k : k+1 \text{ appears left of } k\}$. For $D \subseteq [N-1]$, let $P_D(n) = \{b_1 \le \cdots \le b_N \text{ in } [n] : b_k < b_{k+1}\ \forall k \in D\}$, and

$$F_{N,D}(x_1, \ldots, x_n) = \sum_{b \in P_D(n)} x_{b_1} \cdots x_{b_N}$$

(Gessel's fundamental quasisymmetric function). We write F_α, where α is the composition of N with partial sums D. $C_\sigma := \{w \in [n]^N : \mathrm{std} w = \sigma\}$.

Lemma 5.2 (Gessel's bijection)**.** *For $\sigma \in S_N$ with $D = \mathrm{iDes}\sigma$, the map $w \mapsto b$, $b_k :=$ the letter carrying label k (i.e. $b_k = w_{\sigma^{-1}(k)}$, where $\sigma^{-1}(k)$ is the position of k in σ), is a bijection $C_\sigma \to P_D(n)$ with $x^w = x_{b_1} \cdots x_{b_N}$. Hence $\mathrm{ch}\, C_\sigma = F_{\mathrm{iDes}\sigma}$.*

Proof. Labels increase with letters, so b is weakly increasing. If $b_k = b_{k+1}$, the labels $k, k+1$ sit on equal letters, which are labelled left to right, so k is left of $k+1$ and $k \notin D$. Conversely, given $b \in P_D(n)$, put $w_{\sigma^{-1}(k)} = b_k$. The labels carried by a given value c form an interval $[p, q]$ with $[p, q-1] \cap D = \varnothing$. So in σ the entries $p, \ldots, q$ appear left to right, and std labels the c's exactly that way. Smaller values carry smaller labels, so $\mathrm{std} w = \sigma$. The two maps are inverse. □

Theorem B (components; content Gessel / CMRR)**.** *The connected components of the word quasi-crystal $[n]^N$ are exactly the nonempty C_σ. In b-coordinates $C_\sigma \cong P_D(n)$, and $P_D(n)$ has a unique $\ddot{e}$-source b^0, $b^0_k = 1 + |D \cap [1, k-1]|$, of weight $(\alpha_1, \ldots, \alpha_\ell, 0, \ldots, 0)$. Its character is $F_\alpha(x_1, \ldots, x_n)$. $C_\sigma \neq \varnothing$ iff $\ell(\alpha) \le n$.*

Proof. (1) The operators in b-coordinates. Let $w \in C_\sigma$, let l be the number of letters $\le i$, and suppose w contains an $i+1$, so $b_{l+1} = i+1$. Labels on equal letters increase left to right, so the leftmost $i+1$ carries label $l+1$. *Claim:* w is i-alive iff $l \notin D$ or there is no i (i.e. $b_l \neq i$). If some $i+1$ precedes some i, then the first $i+1$ (label $l+1$) precedes the last i (label l), so $l \in D$. Conversely, $l \in D$ with $b_l = i$ means label $l+1$ (an $i+1$) is left of label l (an i). When alive, $\ddot{e}_i$ turns the leftmost $i+1$ into i. That letter becomes the rightmost i (all i's precede it), so every label is unchanged: std is preserved, and in b-coordinates $\ddot{e}_i$ lowers b_{l+1} from $i+1$ to i. $\ddot{f}_i$ is the inverse partial map. So each C_σ is a union of components. *(2) Connectedness and unique source.* By induction $b \ge b^0$ pointwise for $b \in P_D(n)$. Suppose $b \neq b^0$, let k be minimal with $b_k > b^0_k$, and put $i+1 := b_k \ge 2$. For $j < k$, $b_j = b^0_j \le b^0_k < b_k$, so k is the first index with value $i+1$, i.e. $k = l+1$. If $b_{k-1} = i$, then $k-1 \notin D$, since otherwise $b^0_k = b^0_{k-1} + 1 = b_{k-1} + 1 = b_k$. So by (1), $\ddot{e}_i$ applies (lowering b_k to i stays inside $P_D(n)$) and $\sum b$ drops. Iterating reaches b^0, so $P_D(n)$ is connected. b^0 admits no $\ddot{e}_i$, because the result would be $< b^0$ somewhere, contradicting $b \ge b^0$. So b^0 is the unique $\ddot{e}$-source. Its weight: b^0 takes value j on the j-th block of α. Nonempty iff $b^0_N = 1 + |D| \le n$. *(3)* The character is Lemma 5.2. □

Example 5.3 (C_{21} in $n = 3$)**.** $\sigma = 21$, $D = \{1\}$, $\alpha = (1,1)$: $C_{21} = \{21, 31, 32\}$ with

$$21 \xrightarrow{\ddot{f}_2} 31 \xrightarrow{\ddot{f}_1} 32$$

Source 21 has weight $(1,1,0) = \alpha$ padded. The character is $x_1x_2 + x_1x_3 + x_2x_3 = F_{11} = e_2$. Here 21 is 1-dead and 32 is 2-dead. (Here Q changes nothing relative to the crystal $B(1,1)$. Example 3.6 is the smallest case where it does.)

6. Products: $K_0(\mathcal{Q}_n) \cong \mathrm{QSym}_n$ as rings (Q65)

Proposition 6.1 (the tensor shadow is the product)**.** *For $\sigma_1 \in S_m$, $\sigma_2 \in S_k$: $\{uv : u \in C_{\sigma_1}, v \in C_{\sigma_2}\} = \bigsqcup_{\tau \in \sigma_1 \Cup \sigma_2} C_\tau$, where $\sigma_1 \Cup \sigma_2 = \{\tau \in S_{m+k} : \mathrm{std}(\tau_1 \cdots \tau_m) = \sigma_1,\ \mathrm{std}(\tau_{m+1} \cdots \tau_{m+k}) = \sigma_2\}$. Hence $F_{\mathrm{iDes}\sigma_1} F_{\mathrm{iDes}\sigma_2} = \sum_\tau F_{\mathrm{iDes}\tau}$.*

Proof. $\mathrm{std}(u)$ and $\mathrm{std}(v)$ are the standardisations of the restrictions of $\mathrm{std}(uv)$ to the two position blocks. Conversely, a word w with $\mathrm{std}w = \tau \in \sigma_1 \Cup \sigma_2$ splits as $u\,v$ with $\mathrm{std}u = \sigma_1$, $\mathrm{std}v = \sigma_2$. For characters, use Lemma 5.2 and additivity of weight. □

Example 6.2. $\sigma_1 = 1$, $\sigma_2 = 12$: $\sigma_1 \Cup \sigma_2 = \{123, 213, 312\}$ with iDes $= \varnothing, \{1\}, \{2\}$. So $F_1F_2 = F_3 + F_{12} + F_{21}$, and $C_1 \ddot{\otimes} C_{12} \cong C_{123} \sqcup C_{213} \sqcup C_{312}$.

Definition 6.3. $\mathcal{Q}_n$ is the class of quasi-crystals isomorphic to finite disjoint unions of components of word quasi-crystals $[n]^N$, $N \ge 0$. $K_0(\mathcal{Q}_n)$ is the free abelian group on isoclasses of connected objects, with $[A \sqcup B] = [A] + [B]$, and multiplication $[Q][Q'] = [Q \ddot{\otimes} Q']$. QSym_n is the ring of quasisymmetric polynomials in $x_1, \ldots, x_n$.

Lemma 6.4. $\{F_\alpha(x_1, \ldots, x_n) : \ell(\alpha) \le n\}$ *is a $\mathbb{Z}$-basis of* QSym_n *(in particular, no repeats).*

Proof. $F_\alpha = \sum_{\beta \text{ refines } \alpha} M_\beta$. In n variables $M_\beta = 0$ if $\ell(\beta) > n$. Otherwise M_β has coefficient 1 on $x^{(\beta,0,\cdots)}$, and distinct M_β have disjoint supports, so $\{M_\beta : \ell(\beta) \le n\}$ is a basis. The transition $F \to M$ is unitriangular for refinement. □

Theorem 6.5 (Q65)**.** *(i) $C \mapsto \mathrm{ch}\, C$ is a bijection from isoclasses of connected objects of $\mathcal{Q}_n$ to $\{F_\alpha(x_1..x_n) : \ell(\alpha) \le n\}$.*

(ii) $\mathcal{Q}_n$ is closed under $\ddot{\otimes}$, and $\mathrm{ch}(Q \ddot{\otimes} Q') = \mathrm{ch}\, Q \cdot \mathrm{ch}\, Q'$. Hence $K_0(\mathcal{Q}_n) \cong \mathrm{QSym}_n$ as rings, with isoclasses $\mapsto$ the F-basis.

(iii) (Commutativity for free.) $Q \ddot{\otimes} Q' \cong Q' \ddot{\otimes} Q$ *for* $Q, Q' \in \mathcal{Q}_n$*, although* $\ddot{\otimes}$ *is not symmetric pointwise.*

(iv) (Stability.) Let $\mathrm{res} : \mathcal{Q}_{n+1} \to \mathcal{Q}_n$ *keep* $\{\mathrm{wt}_{n+1} = 0\}$ *and forget index* n*. It commutes with* $\ddot{\otimes}$ *and induces* $x_{n+1} \mapsto 0$*. The inverse limit is* QSym*, with* $[C_\alpha] \mapsto F_\alpha$*.*

Proof. (i) By Theorem B every connected object is $\cong P_D(n)$, and $P_D(n)$ depends only on α. So the isoclass is a function of α, and every α with $\ell(\alpha) \le n$ occurs. Injectivity: the weight of the unique $\ddot{e}$-source is α padded, and isomorphisms preserve sources and weights. (Alternatively, use Lemma 6.4.) (ii) Def. 4.1 is pointwise in the factor data, so $\ddot{\otimes}$ respects isomorphisms and distributes over $\sqcup$. For $Q \subseteq [n]^m$ and $Q' \subseteq [n]^k$ unions of components, Theorem A identifies $Q \ddot{\otimes} Q'$ with $\{uv\} \subseteq [n]^{m+k}$. By Prop. 6.1 this is a union of components. The unit is $\{\varnothing\}$, and associativity comes from concatenation. Weight is additive, so the character is multiplicative. Decomposition into components is canonical, so K_0 is free on connected isoclasses, which by (i) and Lemma 6.4 map bijectively onto a basis. (iii) Both sides lie in $\mathcal{Q}_n$ and have the same character because QSym_n is commutative. By (i) the character determines the isoclass of each component, and multiplicities are read off the F-expansion. (iv) $\{\mathrm{wt}_{n+1} = 0\}$ is closed under $\ddot{e}_i, \ddot{f}_i$ for $i < n$, and on words it is the set of words over $[n]$, so $\mathrm{res}\, C_\sigma^{[n+1]} = C_\sigma^{[n]}$. Weights are ≥ 0, so $\mathrm{wt}_{n+1}(x \ddot{\otimes} y) = 0$ iff both are 0, and the $\ddot{\otimes}$ rule for $i < n$ uses only i-data. Finally $F_\alpha(x_1..x_{n+1})|_{x_{n+1}=0} = F_\alpha(x_1..x_n)$. □

7. Branching: Res_a induces Δ

Theorem 7.1 (Levi branching)**.** *Fix* $1 \le a \le n-1$*, and let* Res_a *forget* $\ddot{e}_a, \ddot{f}_a$*. Read the result as an* $A_{a-1} \times A_{n-a-1}$ *quasi-crystal, with letters* $\le a$ *in the first factor and letters* $> a$ *(shifted down by* a*) in the second. For* $C = C_\sigma$ *with* $D = \mathrm{iDes}\sigma$*,* $|\alpha| = N$*:*

$$\mathrm{Res}_a C_\sigma \cong \bigsqcup_{k=0}^{N} C^{(a)}_{\sigma_1^{(k)}} \boxtimes C^{(n-a)}_{\sigma_2^{(k)}},$$

where $\sigma_1^{(k)} = \mathrm{std}($*values* $1..k$ *of* σ *in position order*$)$ *and* $\sigma_2^{(k)}$ *is the same for the values* $k+1..N$*. Empty terms are dropped.* $\mathrm{iDes}\sigma_1^{(k)} = D \cap [1, k-1]$ *and* $\mathrm{iDes}\sigma_2^{(k)} = (D \cap [k+1, N-1]) - k$*. Hence*

$$\mathrm{ch}\,\mathrm{Res}_a C_\sigma = \sum_k F_{\beta_k}(x_{\le a}) F_{\gamma_k}(x_{>a}) = F_\alpha(x_{\le a}; x_{>a}) = (\Delta F_\alpha)(x_{\le a}; x_{>a}),$$

with $\alpha = \beta_k \cdot \gamma_k$ *if* $k \in D \cup \{0, N\}$ *and* $\alpha = \beta_k \odot \gamma_k$ *(last part of* β_k *merged with first part of* γ_k*) if* $k \notin D$*.*

Proof. Operators with $i \ne a$ never move a letter across $a \leftrightarrow a+1$, so $k = \#\{\text{letters} \le a\}$ is invariant. Fix k and set $\psi(w) = (w|_{\le a},\, w|_{>a} - a)$ (subwords). For $i < a$, the operators and the inversion test see only letters $i, i+1 \le a$, so ψ intertwines them on the first coordinate and fixes the second. The case $i > a$ is symmetric. Since $\mathrm{std} w = \sigma$, the letters $\le a$ sit exactly at the positions of the values $1..k$ of σ, so ψ is injective on the slice. It is surjective onto $C_{\sigma_1} \times C_{\sigma_2}$: put u at those positions and $v + a$ elsewhere. Every letter of u is below every letter of $v + a$, so the standardisation is σ. Each factor is connected (Theorem B), so each slice is one component of the product. For the iDes computation, the relative positions of $j, j+1$ within one value block are inherited from σ. The descent at k itself is lost, and it is recorded in the choice $\cdot$ versus $\odot$. The final equality is the definition of Δ on the F-basis [Grinberg–Reiner 1409.8356, §5], equivalently Gessel's $F_\alpha(x+y)$ formula. □

Example 7.2. $\alpha = (2,1)$, $D = \{2\}$. The cuts $k = 0, 1, 2, 3$ give $(\varnothing, 21)$, $(1, 11)$, $(2, 1)$, $(21, \varnothing)$, so $\Delta F_{21} = 1 \otimes F_{21} + F_1 \otimes F_{11} + F_2 \otimes F_1 + F_{21} \otimes 1$.

Corollary 7.3. $\mathrm{Res}_a(Q \ddot{\otimes} Q') = \mathrm{Res}_a Q \ddot{\otimes} \mathrm{Res}_a Q'$, *and* $(A \boxtimes B) \ddot{\otimes} (A' \boxtimes B') \cong (A \ddot{\otimes} A') \boxtimes (B \ddot{\otimes} B')$. *Hence* Res_a *is a ring map* $K_0(\mathcal{Q}_n) \to K_0(\mathcal{Q}_a) \otimes K_0(\mathcal{Q}_{n-a})$, $f \mapsto f(x_{\le a}; x_{>a})$. *As* $a, n-a \to \infty$ *this is* $\Delta : \mathrm{QSym} \to \mathrm{QSym} \otimes \mathrm{QSym}$, *and its multiplicativity (the bialgebra axiom) is realised by quasi-crystals.*

Proof. The $\ddot{\otimes}$ rule is per index. For $i < a$ all the i-data of (a, b) are those of a, so $((a, b), (a', b')) \mapsto ((a, a'), (b, b'))$ is the isomorphism. □

8. Crystals → quasi-crystals: $\mathrm{Sym}_n \hookrightarrow \mathrm{QSym}_n$

Corollary 8.1. *(a)* $Q_{B(\lambda)} \in \mathcal{Q}_n$ *and* $\mathrm{ch}\, Q_{B(\lambda)} = s_\lambda$.

(b) $[B] \mapsto [Q_B]$ *is an injective ring map* $\mathrm{Sym}_n = K_0(\textit{polynomial } \mathfrak{gl}_n \textit{ crystals}) \to K_0(\mathcal{Q}_n) = \mathrm{QSym}_n$. *Under* ch *it is the inclusion* $\mathrm{Sym}_n \subset \mathrm{QSym}_n$.

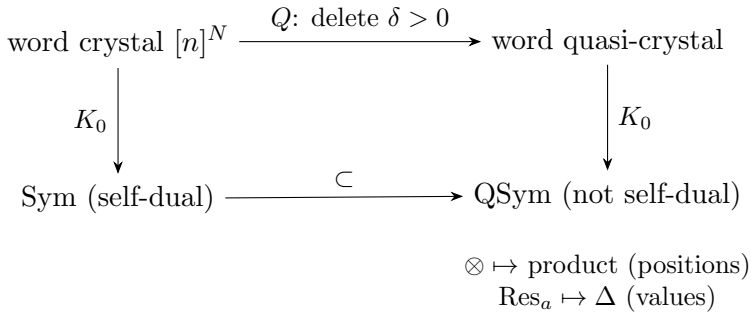
(c) $[Q_{B(\lambda)}] = \sum_{T \in SYT(\lambda)} [C_{\mathrm{Des}T}]$, *so* Q *splits* $B(\lambda)$ *into* f^λ *quasi-components (CMRR2 Cor. 4.10).*

(d) $\mathrm{Res}_a \circ Q = Q \circ \mathrm{Res}_a$.

Proof. (a) Black box 2 realises $B(\lambda)$ as a component of $[n]^N$. Quasification is pointwise, so Q of a subcrystal is the restriction of $Q_{[n]^N}$, which is the word quasi-crystal (Lemma 3.5(e)). The restriction is closed under $\ddot{e}, \ddot{f}$, hence is a union of quasi-components. The set and weights are unchanged. (b) Additivity is clear. Multiplicativity is Theorem 4.3 (Black box 1). $\mathrm{ch}\, Q_B = \mathrm{ch}\, B$, so the map is the inclusion. (c) Theorem 6.5(i) forces $[Q_{B(\lambda)}] = \mathrm{ch}^{-1}(s_\lambda)$. Now expand by Black box 3. (d) Def. 3.3 is per index. □

Example 8.2. In $\mathfrak{gl}_2$, $B(2,1)$ is the 2-element crystal of Example 3.6, and $s_{21}(x_1, x_2) = F_{12} + F_{21}$: two quasi-components. In $\mathfrak{gl}_3$, $s_{21} = F_{12} + F_{21}$ again, and the $x_1 x_2 x_3$ coefficient is $2 = 1 + 1$.

9. Punchline: tensor ↔ product, branching ↔ coproduct



(1) **Two operations, not one.** Seed Path 4 asks why the Sym coproduct matches both the crystal tensor rule and restriction. It doesn't match both. Tensor products shadow the *product* (Prop. 6.1, the restriction of σ to *position* blocks). Branching shadows the *coproduct* (Theorem 7.1, the restriction of σ to *value* blocks). In Sym both give $c^\lambda_{\mu\nu}$ because Sym is self-dual. In QSym they separate. Minimal separator: C_{312} (character F_{21}) is a component of $C_1 \ddot{\otimes} C_{12}$, yet $F_1 \otimes F_2$ does not occur in ΔF_{21}.

(2) **Zero-cross-cut-bracket locus.** By the defect identity, quasification is the locus with no bracket inside either factor and none across the cut. The crystal keeps the cross-cut bracket and pays with a non-strict (bracketed) tensor rule. The quasi-crystal deletes it, which makes the tensor rule factor-local for every cut, and pays with self-duality.

(3) **Positions ↔ values.** These two restrictions are swapped by $\sigma \leftrightarrow \sigma^{-1}$. The components C_σ remember all of σ, so they live at the level of FQSym, whose self-duality $F_\sigma \leftrightarrow G_{\sigma^{-1}}$

(Malvenuto–Reutenauer) is the plausible home of this swap. *Grade: hunch, interpretive, not proved here.*

(4) **Seed Q1 / Q4.** QSym is the terminal combinatorial Hopf algebra (Aguiar–Bergeron–Sottile). Its "$q = 0$" categorical avatar is a crystal-like graph, the type-A quasi-crystal, just as $B(\lambda)$ is for $\mathrm{Sym} = K_0(\mathrm{Rep}\, U_q(\mathfrak{gl}_\infty))$. The inclusion $\mathrm{Sym} \subset \mathrm{QSym}$ is the K_0-shadow of the forgetful functor Q, i.e. of "a crystal is a union of quasi-crystals". This is a bialgebra-level statement only. The antipode and the ABS character ζ have *not* been given categorical meaning.

10. Who owns what

- Word quasi-crystals and quasi-Kashiwara operators: Cain–Malheiro (hypoplactic monoid), in the convention of Cain–Malheiro–F. Rodrigues–I. Rodrigues [CMRR1 2309.14887 Def. 2.13]. Quasification: [CMRR2 2309.14898 Def. 4.5]. The f^λ count: [CMRR2 Cor. 4.10].
- Quasi-tensor product and Theorem A: Cain–Guilherme–Malheiro [2301.00271 Thm 5.1, Prop. 5.9, Def. 6.9], with word reversal.
- Theorem B, Prop. 6.1, Theorem 7.1 (content): Gessel 1984; Malvenuto–Reutenauer 1995 (shuffle product, $F_\alpha(x+y)$); Krob–Thibon (quasi-symmetric functions and 0-Hecke algebras).
- The defect identity, Theorem 4.3, and the K_0/ring/branching packaging of Theorem 6.5: we did not find them stated in Maas-Gariépy 2302.07694, Brauner–Corteel–Daugherty–Schilling 2503.14782, CMRR 2309.14887/2309.14898, Brauner–Daugherty–Mason–Schilling 2607.12232, or Kobayashi 2512.06273. This check covers only these sources, and the absence of a statement there is not evidence of novelty. Given how short the proofs are, we regard the content as folklore at the level of Gessel / Malvenuto–Reutenauer / Krob–Thibon, and it should be cited that way.

11. Open questions

(1) **Intrinsic characterisation.** $\mathcal{Q}_n$ is defined via words. Do CMRR's local axioms for quasi-crystals (2309.14898) characterise the objects of $\mathcal{Q}_n$? A yes makes Theorem 6.5 axiomatic. A no would be an axiom-satisfying quasi-crystal not isomorphic to any $P_D(n)$.

(2) **Antipode and ζ.** Is there a categorical meaning (a duality on $\mathcal{Q}_n$, a "sign" or reversal functor) inducing the antipode of QSym, or the ABS character ζ? Partial: none.

(3) **Cross-cut brackets and self-duality.** Does the cross-cut bracket "cause" the self-duality of Sym (RSK)? No separator has been proposed. This is interpretive.

(4) **Other kill rules (closed, empty).** The opposite kill clause ($\ddot{\varphi}(x) > 0 \wedge \ddot{\varepsilon}(y) > 0$), with the operator choices mirrored, is our rule applied to the swapped pair (y, x). That is the opposite monoidal structure $Q \mathbin{\ddot{\otimes}^{\mathrm{op}}} Q' := Q' \mathbin{\ddot{\otimes}} Q$. Its K_0 is the same commutative ring. So the swapped-order control (which fails) shows a *different* rule, not a different category.

Appendix A. Verification

Every displayed example was re-implemented from the printed statements (`proofs/prove277/examples.py`, independent of the PROVE 275/276 code). Results: Theorem A rule versus the definition, 1272 cases ($n \le 4$, $|u| + |v| \le 3$), 0 failures; swapped kill clause, 6 failures on $n = 2$, $|u| = |v| = 2$ (the control breaks); defect identity on words ($n = 3$, $|u| = |v| = 3$), 0 failures; components = std classes ($n = 3$, $N = 4$); C_{21} chain; 211/212 dead; $F_1F_2 = F_3 + F_{12} + F_{21}$ ($n = 4$); the shuffle set $\{123, 213, 312\}$; ΔF_{21} cuts; $\det^{-1} \otimes \det$ and $B(0,-1) \otimes B(2,1)$ as stated; Theorem B unique source of weight α padded ($N \le 4$, $n \le 4$, all σ), 0 failures; Theorem 7.1 (ψ a bijection onto $C_{\sigma_1} \times C_{\sigma_2}$ and the iDes formulas) on 464 slices,

0 failures. Earlier larger runs: `proofs/prove275/`, `proofs/prove276/` (Q66: 134 tensor pairs, 5418 elements, 0 failures; control 4500 failures).